Relational Time Geometry:
Thermodynamic Floors, Geometric Noise, and Spectral Impedance in Memory-Dependent Stability

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Abstract

We develop an *operational* definition of time as a relational quantity emerging from an observer's enforcement of a geometric constraint manifold ("The Law") on dynamical states using a history-dependent memory operator. Across the RTG program we recover three roles of memory: (1) **Drag** in continuous fractional flows (validated escape scaling $\gamma_{\text{esc}} = 1/(2\alpha)$), (2) **Friction** in stochastic kinetics (validated barrier exponent $p = 3/2$), and (3) **Inertial Momentum** in discrete finite-memory maps (validated fracture clocks $\tau_{\text{exit}} \approx kM + c$).

The main empirical result is a dense chaotic sweep of an anchored logistic source, showing that the steady-state maintenance power saturates at a robust **Thermodynamic Floor** P_{floor}. In deep chaos ($\lambda > 0.15$), P_{floor} follows an affine equation of state $P_{\text{floor}}(\lambda) \approx \gamma\,\lambda + E_0$ ($R^2 \approx 0.88$, $N = 42$ non-resonant points). A bootstrap analysis yields $\gamma \approx 0.227$ (95% CI $\approx [0.20, 0.26]$) and $E_0 \approx 0.089$ (95% CI $\approx [0.075, 0.10]$). We identify the positive intercept as a **Geometric Noise Floor**, numerically consistent with the intrinsic variance of a state confined to the unit interval ($1/12 \approx 0.083$). The slope γ has units of Energy/Nat and quantifies a metabolic cost of screening information production. We further characterize **Resonance Windows** as regions of high **Spectral Impedance**, where the memory kernel's transfer function aligns with the orbit spectrum. Finally, we give a constructive (conjectural) bridge to quantum measurement: among projection maps, Born statistics may be selected by a *Principle of Least Projection Energy*.

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1 Introduction

In canonical quantum mechanics, time is an external parameter; in general relativity, it is dynamical but local. The *relational time* hypothesis instead treats time as an emergent quantity derived from correlations between subsystems. **RTG** operationalizes this idea via a minimal control-theoretic setup: a *Relational Observer* enforces a geometric constraint ("The Law") on a dynamical source using a history-dependent memory operator.

This manuscript integrates the three phases of the RTG program (Flow, Noise, Map) into a single narrative. The central question is energetic:

> *What is the energetic cost of maintaining simultaneity (containment) in an unstable or chaotic universe?*

Our main finding is that, in the discrete-chaotic sector, the cost does *not* vanish with increasing memory depth. Instead it saturates at a nonzero thermodynamic floor composed of a geometric baseline and an entropy-dependent term.

1.1 Notation summary

- $\alpha \in (0, 1]$: fractional memory order.
- $M \in \mathbb{N}$: discrete memory depth.
- $r \in [0, 4]$: logistic nonlinearity parameter.
- $\lambda(r)$: Lyapunov exponent (nats/step).
- $\Omega = [0, 1]$: operational constraint manifold ("The Law").
- $P_{\text{Law}}(M)$: steady-state maintenance power (mean squared projection).
- $P_{\text{floor}}(r)$: floor estimate (median of $P_{\text{Law}}(M)$ across sampled M).
- $G_M(\alpha, M)$: inertial gain $\sum_{j=0}^{M-1} w_j^{(\alpha)}$.

1.2 Epistemic status

We distinguish between:

- **Validated results:** supported by numerical sweeps and statistical checks reported here (fracture clocks, the floor equation of state, and the resonance audits).
- **Recovered pillars:** results established in earlier phases of the RTG program and summarized here for synthesis (drag and friction targets).
- **Constructive proposals:** mechanisms offered as mathematically explicit bridges but not yet derived from first principles (Born-rule variational selection).
- **Conjectures:** hypotheses intended to motivate future tests (the κ_3 sector / rank-3 obstruction).

2 Mathematical foundations: Curvature rank and ghosts

A recurring theme in the RTG program is that long-lived metastable "ghost" structures arise when low-order curvature cancels and the first non-vanishing term occurs at higher order. One way to formalize this is via a *curvature rank* (a "peeling" diagnostic) familiar from bifurcation theory [1, 2].

Definition 2.1 (Curvature rank)**.** *Let $f(x;\mu)$ be a smooth map or flow near a critical point (x^*, μ_c). The curvature rank is*

$$R(f, x^*) := \min\{k \geq 1 : \ \partial_x^{(k)} f(x^*; \mu_c) \neq 0\}.$$

Higher rank implies flatter local geometry and longer bottlenecks. In later sections, discrete finite-memory updates convert such bottlenecks into deterministic fracture events.

3 Methods: The RTG minimal model and estimation protocol

3.1 Source dynamics: logistic map

We use the logistic map as a prototypical generator of instability [3, 4]:

$$x_{n+1} = f_r(x_n) := r x_n(1 - x_n), \qquad x_n \in [0, 1]. \tag{1}$$

Chaotic instability is quantified by the Lyapunov exponent (in nats/step) $\lambda(r)$, with $\lambda_{\max} = \ln 2$ at $r = 4$.

3.2 Anchored finite-memory Caputo-increment map

We adopt an anchored finite-memory fractional integral in the *Caputo-increment* form [5]. Define increments $\Delta x_n := f_r(x_n) - x_n$. For memory order $\alpha \in (0, 1]$ and depth M, define the power-difference weights

$$w_j^{(\alpha)} := \frac{(j+1)^\alpha - j^\alpha}{\Gamma(1+\alpha)}, \qquad j = 0, 1, \ldots, M-1. \tag{2}$$

The *natural* (unconstrained) update is

$$x_{n+1}^{\text{nat}} = x_0 + \sum_{j=0}^{\min(n, M-1)} w_j^{(\alpha)} \, \Delta x_{n-j}. \tag{3}$$

3.3 The Law and greedy projection

The Law is the bounded manifold $\Omega = [0, 1]$. The observer applies a greedy projection $\Pi_\Omega : \mathbb{R} \to [0, 1]$:

$$x_{n+1} := \Pi_\Omega(x_{n+1}^{\text{nat}}). \tag{4}$$

The **operational law energy** over a measurement window T is

$$E_{\text{proj}}(M; T) := \sum_{n=0}^{T-1} |x_{n+1} - x_{n+1}^{\text{nat}}|^2, \tag{5}$$

and the **maintenance power** is

$$P_{\text{Law}}(M) := \frac{E_{\text{proj}}(M; T)}{T}. \tag{6}$$

3.4 Inertial gain

A key control parameter is the **memory gain**

$$G_M(\alpha, M) := \sum_{j=0}^{M-1} w_j^{(\alpha)} \sim \frac{M^\alpha}{\Gamma(1+\alpha)}. \tag{7}$$

For $\alpha = 0.98$ and $M = 300$, $G_M \approx 270$, so the history term acts as a very high-gain inertial "kick."

3.5 Floor estimation and resonance flagging

For each r, we estimate a floor $P_{\text{floor}}(r)$ as the *median* of $P_{\text{Law}}(M)$ over a fixed set of memory depths (here $M \in \{60, 100, 200, 300\}$). Resonance windows are detected operationally by the *spread* of $P_{\text{Law}}(M)$ across this set. We flag a parameter value as resonant when the mean inter-M spread exceeds a fixed threshold (here 0.4 in normalized units; see Appendix C for a spectral proxy that predicts this spread).

4 Recovered pillar I: Continuous dynamics (memory as drag)

For fractional flows $D_t^\alpha x = f(x)$, discretization introduces a factor h^α that renormalizes the memory operator [6, 7]. Memory suppresses instability and acts as **drag**. In the Kramers escape setting [8, 9], this yields a modified scaling

$$\tau_{\text{esc}}(\mu) \sim \mu^{-1/(2\alpha)}. \tag{8}$$

Table 1: Target 2 validation (Drag): escape-time scaling exponent $\gamma_{\text{esc}}(\alpha)$.

α	$\gamma_{\text{esc,fit}}$	$\gamma_{\text{esc,theory}} = 1/(2\alpha)$	Rel. Error (%)
1.00	0.500	0.500	0.0
0.95	0.526	0.526	0.0
0.90	0.556	0.556	0.0
0.80	0.632	0.625	+1.1
0.70	0.800	0.714	+12.0

5 Recovered pillar II: Stochastic dynamics (memory as friction)

In stochastic systems, memory alters kinetics but preserves barrier geometry. Using anomalous diffusion models [10] and survival analysis [11], we fit escape times as

$$\mathbb{E}[\tau_{\text{esc}}] \sim \exp(C\,\mu^p/\sigma^2), \tag{9}$$

and find the cubic-barrier exponent $p = 3/2$ is robust (Target 3).

Table 2: Target 3 validation (Friction): Kaplan–Meier analysis favors the invariant barrier exponent $p = 3/2$ over an α-dependent alternative.

α	R_A^2 $(p = 3/2)$	R_B^2 $(p = 3/2\alpha)$	ΔR^2	censor_frac
1.00	0.930	0.930	0.000	0.33
0.95	0.893	0.886	+0.007	0.30
0.90	0.869	0.854	+0.015	0.24
0.80	0.779	0.742	+0.037	0.08
0.70	0.851	0.802	+0.049	0.002
0.60	0.821	0.750	+0.070	0.00

6 Recovered pillar III: Discrete dynamics (memory as inertial momentum)

In discrete maps ($h = 1$), memory is unrenormalized and acts as **inertial momentum**. Finite memory induces deterministic **fracture**: the natural update (3) can overshoot the domain Ω, and greedy projection imposes repeated corrections. Empirically, the **fracture time** scales approximately linearly with memory depth,

$$\tau_{\text{exit}}(M) \approx kM + c, \qquad k \in \{1, 2\}. \tag{10}$$

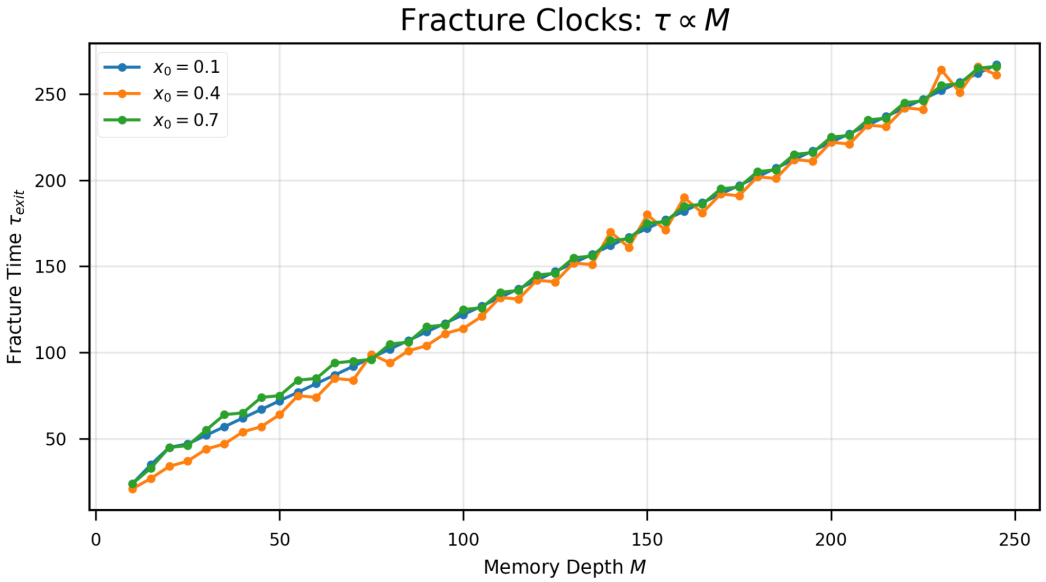


Figure 1: **Fracture clocks.** Exit time τ_{exit} versus memory depth M for several initial conditions. The near-linear scaling supports the inertial-momentum mechanism.

6.1 The period-2 bypass phenomenon (mechanistic note)

At $\alpha \approx 1$ and large M, we observe restructuring of the classical period-doubling cascade: certain low-period orbits (notably period-2) may be "bypassed" because the transient memory sum accumulates a ballistic kick before long-time cancellation occurs.

Mechanism. The period-2 orbit alternates sign (Δx_n changes sign each step). While the infinite-history sum can converge by cancellation, the transient sum over the first $\sim M$ steps accumulates a ballistic kick of order $\sim \epsilon G_M$ (cf. Theorem 7.1), which fractures the domain before steady-state cancellation is reached.

7 The inertial momentum theorem

Theorem 7.1 (Inertial Momentum Bound)**.** *Assume* $\Delta x_n \geq \epsilon > 0$ *over a window of length* M*, i.e. for* $n \in \{N_0, \ldots, N_0 + M - 1\}$*. Then*

$$x^{\text{nat}}_{N_0+M} - x_0 \geq \epsilon \sum_{j=0}^{M-1} w_j^{(\alpha)} = \epsilon\, G_M(\alpha, M) \sim \epsilon \frac{M^\alpha}{\Gamma(1+\alpha)}. \tag{11}$$

Proof sketch. Using the power-difference weights (2),

$$\sum_{j=0}^{M-1} w_j^{(\alpha)} = \frac{1}{\Gamma(1+\alpha)} \sum_{j=0}^{M-1} \left[(j+1)^\alpha - j^\alpha\right] = \frac{M^\alpha}{\Gamma(1+\alpha)},$$

a telescoping sum. The lower bound follows by summing $\epsilon\, w_j^{(\alpha)}$ over the window. □

This explicit growth explains why linear fracture clocks (10) are generic near $\alpha \approx 1$: finite memory provides a sufficient mechanism for finite-time exit.

8 Thermodynamics: the floor and equation of state

8.1 From scaling hypothesis to floor

A natural early hypothesis was that deeper memory would recover the Law asymptotically, e.g. $P_{\text{Law}}(M) \sim M^{-\beta}$ with $\beta > 0$. Dense sweeps in the deep-chaotic regime instead show that $P_{\text{Law}}(M)$ does *not* decay to zero as $M \to \infty$. It saturates at a robust floor P_{floor}.

8.2 Equation of state

Restricting to non-resonant points (flagged "OK") with $\lambda > 0.15$ (deep chaos, $N = 42$), we find a clean affine relationship:

$$P_{\text{floor}}(\lambda) \approx \gamma\, \lambda + E_0, \tag{12}$$

with $R^2 \approx 0.88$. A bootstrap analysis yields approximately:

- **Slope (metabolic efficiency):** $\gamma \approx 0.227$ (95% CI $\approx [0.20, 0.26]$).
- **Intercept (baseline floor):** $E_0 \approx 0.089$ (95% CI $\approx [0.075, 0.10]$).

Here λ is measured in *nats per step*. To convert to bits/step, scale by $1/\ln 2$.

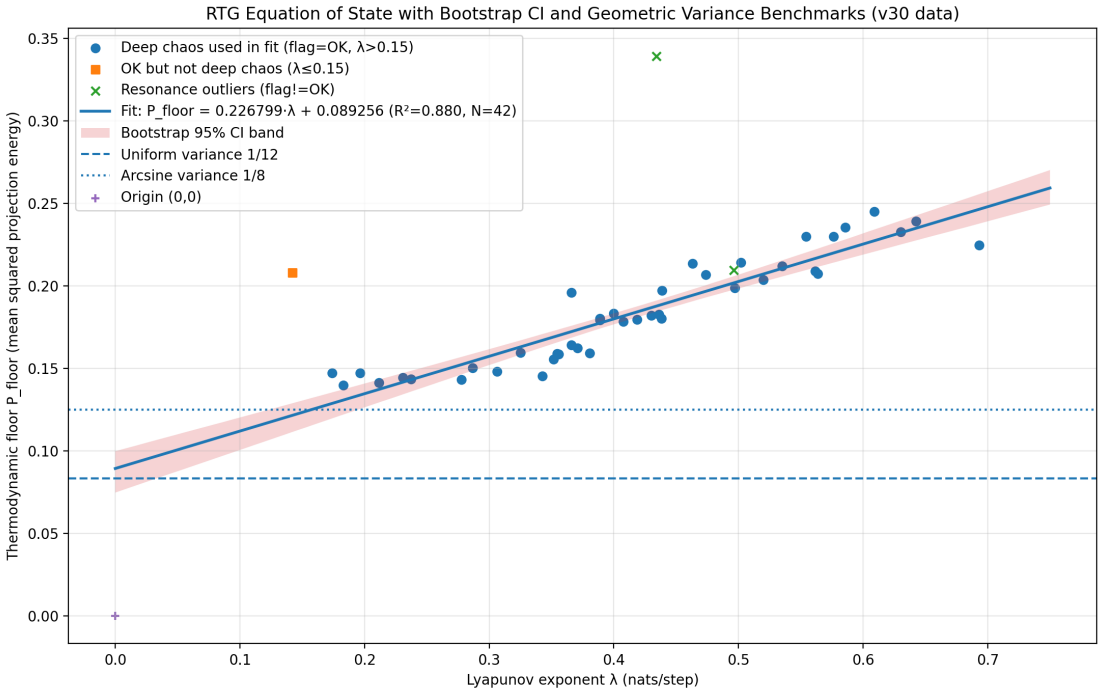


Figure 2: **Thermodynamic equation of state for RTG (v30 sweep).** Scatter: P_{floor} vs. λ for non-resonant deep-chaos points ($N = 42$). Solid line: best-fit affine law (12). Shaded region: 95% bootstrap confidence band. Horizontal dotted lines: reference variances $1/12$ (uniform on $[0, 1]$) and $1/8$ (arcsine invariant density at $r = 4$). The CI band excludes the origin, supporting a nonzero baseline $E_0 > 0$.

8.3 Interpretation I: the geometric noise floor E_0

The positive intercept E_0 is a **static cost of observation**. Even as $\lambda \to 0$, the high-gain inertial operator repeatedly overshoots the constraint and is corrected by projection. Empirically, E_0 is close to the intrinsic variance of a bounded state:

$$\text{Var}(\text{Unif}[0, 1]) = \frac{1}{12} \approx 0.083, \qquad \text{Var}(\text{arcsine at } r = 4) = \frac{1}{8} = 0.125.$$

Remark 8.1 (Bracketing the baseline)**.** *The fitted intercept $E_0 \approx 0.089$ lies between $1/12$ and $1/8$. This suggests that, under large inertial gain, the observer effectively pays the variance cost of containing a near-maximal-entropy state inside Ω, interpolating between "uniform" and "fully chaotic" reference measures.*

8.4 Interpretation II: the slope γ as metabolic efficiency

While E_0 represents the baseline geometric cost, the slope γ represents the **dynamic cost of information erasure**.

Conceptually, the observer corrects state deviations only after they have accumulated across an effective memory horizon. In a chaotic interval, a typical separation grows as $\delta x(t) \sim e^{\lambda t}$. Over a correction lag τ_{eff} this yields an accumulated "ballistic" overshoot. Projection must dissipate that overshoot to enforce Ω, so larger λ demands larger average correction energy. The observed linearity suggests that, within the explored sector, this screening operates in an approximate linear-response regime with respect to information production. Over an effective correction lag τ_{eff}, an uncorrected perturbation grows as $e^{\lambda \tau_{\text{eff}}}$, so larger λ implies faster divergence accumulation and larger projection work per step.

Dimensional consistency. The slope has dimensions

$$[\gamma] = [P_{\text{floor}}]/[\lambda] = [\text{Energy/step}]/[\text{nats/step}] = [\text{Energy/nat}],$$

which is dimensionally analogous to information-thermodynamic bounds such as Landauer's cost for erasure [12, 13]. In the RTG setting, the "energy" is geometric (projection work) and the constraint manifold acts as the information sink, so the analogy is structural rather than literal.

Comparison to thermal scales (order-of-magnitude). If we identify the RTG geometric energy scale with an effective thermal scale $k_B T$, then $\gamma \approx 0.23$ corresponds to $\sim 0.23\, k_B T$ per nat screened. For comparison, Landauer's bound is $k_B T \ln 2 \approx 0.69\, k_B T$ per bit erased [12, 13]. This comparison is only dimensional, but it places the RTG slope in a physically plausible range.

Control-theoretic context. Invariance-entropy results in nonlinear control link the information rate required to maintain a set to dynamical instability [14, 15]. The empirical linearity in Figure 2 suggests that, over the explored chaotic sector, the RTG observer operates in an approximate linear-response regime with respect to information production.

8.5 Resonance as spectral impedance

The affine law (12) is punctuated by **resonance windows** (outliers flagged "RES") where P_{floor} becomes strongly memory-dependent. Operationally, resonance manifests as an anomalously large spread of $P_{\text{Law}}(M)$ across sampled memory depths.

We interpret these windows as regions of high **spectral impedance**: dominant frequencies of the orbit align with the poles of the memory kernel transfer function, amplifying effective gain. Prominent resonance clusters occur near periodic windows of the logistic map, especially the period-3 window around $r \approx 3.83$ [16, 4].

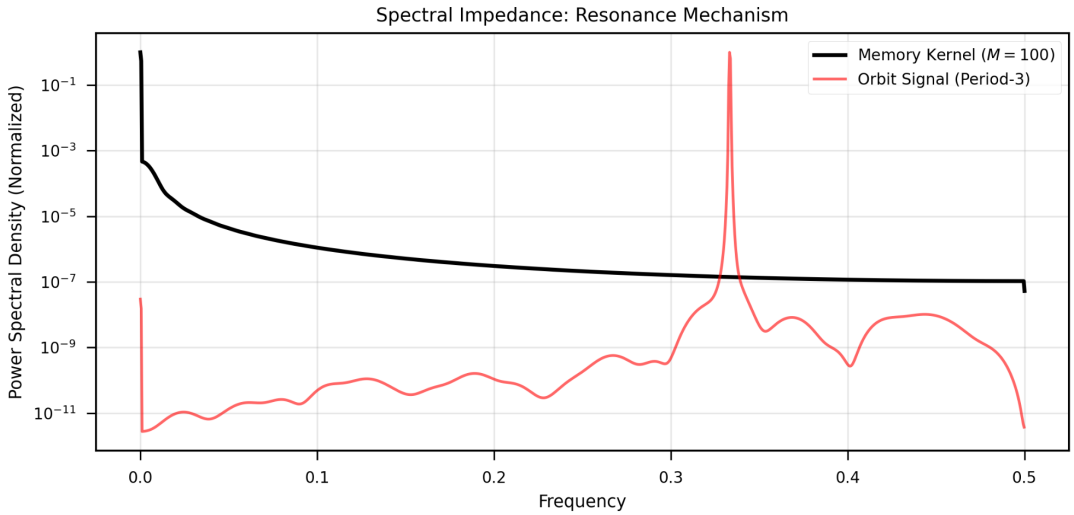


Figure 3: **Spectral impedance mechanism.** Example power spectra (normalized): memory-kernel response vs. orbit increment spectrum in a period-3 window. Resonance corresponds to strong spectral overlap, amplifying projection work.

8.6 Program synthesis: renormalization of memory gain

The three pillars (Drag/Friction/Momentum) and the thermodynamic floor can be organized under a single principle:

Whether the effective memory gain is renormalized or unrenormalized.

In continuous flows, discretization introduces an h^α factor that keeps gain finite as $h \to 0$ (drag). In stochastic kinetics, memory modifies prefactors while preserving barrier geometry (friction). In discrete maps ($h = 1$), gain grows as $G_M \sim M^\alpha$ and creates inertial overshoot, producing a nonzero floor and fracture clocks.

9 Measurement and emergent statistics (constructive proposal)

9.1 Jacobian construction

Let an internal phase variable $z \in [0, 1]$ evolve ergodically with invariant density $\pi(z)$. Define cumulative distributions

$$H(z) = \int_0^z \pi(u)\, du, \tag{13}$$

$$F(x) = \int_{-\infty}^{x} |\psi(x')|^2\, dx'. \tag{14}$$

Then the map $G(z) = F^{-1}(H(z))$ transforms internal measure to the Born measure:

$$\rho(x) = \frac{\pi(z)}{|G'(z)|}\bigg|_{z=G^{-1}(x)} = |\psi(x)|^2.$$

This is a constructive measure-theoretic route to Born statistics [17, 18].

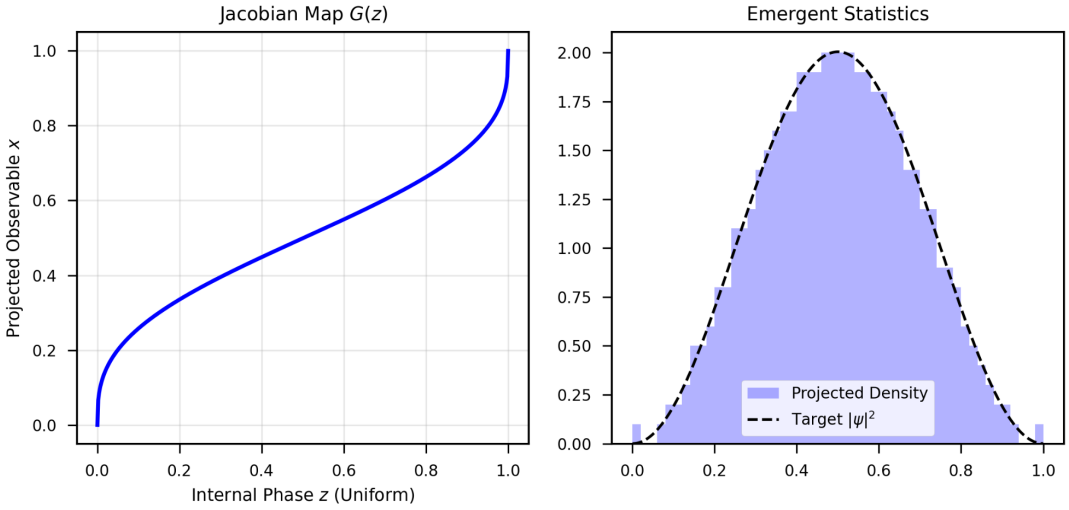


Figure 4: **Constructive Born rule.** (Left) A Jacobian map $G(z)$ transporting a uniform internal phase z to an observable coordinate x. (Right) The induced histogram matches the target density $|\psi|^2$ in this example.

9.2 Energetic selection principle (conjecture)

Conjecture 9.1 (Principle of Least Projection Energy)**.** *Among admissible statistical maps G, the physically realized probability rule is selected by minimizing the maintenance power P_{Law} (or its floor P_{floor}) subject to the dynamical constraints of the source.*

Remark 9.2 (Falsification criterion for Conjecture 9.1)**.** *If alternative statistical rules (e.g. $\rho(x) \propto |\psi(x)|^{2+\varepsilon}$) yield* lower *maintenance power than the Born rule in RTG simulations under comparable constraints, then Conjecture 9.1 is disfavored.*

10 Algebraic origin (conjectural): the κ_3 sector

A long-term RTG conjecture is that the energetic obstruction underlying discrete floors is an algebraic (non-Abelian) curvature of rank three. Operationally, rank-3 curvature can appear flat to low-order probes but still force a nonzero correction at third order, analogous to nested commutators in the Baker–Campbell–Hausdorff expansion [19, 20, 21, 22].

10.1 Frustrated triangle example

Example (Frustrated Triangle). Consider SU(2) edge holonomies on a triangle with rotations by an angle θ around axes separated by 120°:

- $\mathcal{R}_1 = 0$: linear terms cancel by $\mathbb{Z}_3$ symmetry.
- $\mathcal{R}_2 = 0$: pairwise commutators cancel.
- $\mathcal{R}_3 > 0$: the nested commutator $[X_1, [X_2, X_3]]$ survives.

This configuration appears flat to low-order diagnostics but contains hidden rank-3 curvature—an algebraic prototype for a "ghost manifold" that is stable until a higher-order probe (or finite-memory truncation) exposes the obstruction.

10.2 Algebraic depth vs. time depth

Remark 10.1 (Time depth $\neq$ algebraic depth)**.** *Even for very large M, the RTG memory operator is a* linear *functional of past increments. Thus it can be deep in time but shallow in algebraic order. If the underlying obstruction is genuinely rank-3, increasing M alone may not remove the baseline cost unless the observer's update rule also increases in algebraic order.*

11 Experimental protocol and falsification

11.1 Memory engineering in open quantum systems

We propose a controllable non-Markovian platform (e.g. superconducting qubits) where an effective memory depth M can be tuned. Two generic approaches are:

1. **Ancilla register:** couple the system to M ancilla qubits via partial-SWAP interactions to implement a programmable memory kernel.
2. **Pulse shaping / dynamical decoupling:** use pulse sequences whose inter-pulse spacing encodes history dependence.

11.2 Expected effect size

Using the v30 regression, $P_{\text{floor}} \approx 0.227\,\lambda + 0.089$ (normalized units). For a system engineered with $\lambda \approx 0.5$ (moderate chaos), the predicted floor is $P_{\text{floor}} \approx 0.20$. Resonant outliers are expected near periodic windows [4]:

- Period-3: $r \approx 3.83$ (strongest resonance in our audit).
- Period-5: $r \approx 3.74$.
- Period-6: $r \approx 3.63$.

(See Appendix C.)

11.3 Falsification criteria

The RTG fracture-clock and floor hypotheses are disfavored if:

1. **No staircase/plateaus:** $\tau_{\text{exit}}(M)$ is well-fit by a smooth law (exponential, power) with no statistically significant linear plateaus.
2. **Clock invariance fails:** data does not collapse when plotted against discrete depth $M = \lfloor \tau_B / \Delta t \rfloor$.
3. **No resonance coupling:** spectral-overlap proxies fail to predict enhanced spread in $P_{\text{Law}}(M)$ near periodic windows.
4. **Intercept collapses:** the floor equation of state fails to reproduce a positive intercept $E_0 > 0$ under controlled variations of protocol and kernel (warmup, sampling, projection rule).

12 Limitations

1. **Single map family:** The thermodynamic floor is validated here primarily for the logistic map. Universality across other unimodal maps remains to be tested.
2. **Fixed α:** The dense sweep uses $\alpha = 0.98$. Floor behavior at smaller α (stronger memory) is largely unexplored.
3. **Greedy projection:** Results depend on Π_Ω. Alternative constraints or optimal control schemes may modify the floor.
4. **Resonance characterization:** The spectral-impedance interpretation is supported by proxies but not yet a full predictive theory.
5. **Constructive quantum bridge:** The Born-rule section is a proposal and does not establish unitarity or full quantum dynamics.

13 Summary of results

Table 3: Summary of validated results and conjectural extensions across the RTG program.

Sector	Observable	Result	Evidence
Continuous (Drag)	Escape scaling	$\gamma_{\text{esc}} = 1/(2\alpha)$	Table 1
Stochastic (Friction)	Barrier exponent	$p = 3/2$ invariant	Table 2
Discrete (Momentum)	Fracture clock	$\tau_{\text{exit}} \approx kM + c$	Figure 1 and Theorem 7.1
Discrete (Floor)	Equation of state	$P_{\text{floor}} = \gamma\lambda + E_0$	Figure 2
Conjectural / under development:			
Born rule	Variational selection	Least projection energy	Conjecture 9.1 and Figure 4
κ_3 sector	Algebraic origin	Rank-3 obstruction	Section 10

14 Open questions

1. $M \to \infty$ **limit:** Does the accessible "survival" dynamics converge under increasing memory depth, or do new fracture branches appear?

2. **Floor universality:** Is the affine equation of state (12) universal across unimodal map families (tent, sine), or specific to the logistic map?

3. **Strong-memory regime:** For $\alpha \lesssim 0.7$, does a distinct universality class emerge in fracture and floor behavior?

4. **Continuous limit:** In discretized fractional flows, the renormalized gain $h^\alpha \sum_j w_j \sim T^\alpha$ stays finite as $h \to 0$. If gain remains bounded and the constraint is smooth, does the overshoot–projection cycle become infinitesimal, implying $E_0 \to 0$ in the continuous limit? This would distinguish the discrete floor as a discretization-dependent phenomenon.

5. **Born rule selection:** Does RTG projection dynamics naturally select the Jacobian map, or is additional structure required to single out $|\psi|^2$?

6. **Resonance prediction:** Can the spectral overlap proxy quantitatively predict resonance *amplitude*, not just location?

7. **Algebraic vs. temporal depth:** How can an observer increase algebraic order (beyond linear memory) and does this reduce the baseline floor?

15 Conclusion

This work began with the hypothesis that deep memory could recover a perfect Law of simultaneity from unstable dynamics. Instead, the discrete-chaotic sector reveals a more thermodynamic reality:

- **Inertia:** unrenormalized memory creates gain $G_M \sim M^\alpha$ that drives overshoot and finite-time fracture.
- **Geometry:** the baseline cost is a geometric noise floor consistent with bounded-state variance ($E_0 \approx 1/12$).
- **Entropy:** the dynamic cost scales with information production ($P_{\text{floor}} \propto \lambda$) with slope γ interpretable as Energy/Nat.

In RTG, relational time is not free: it is maintained by finite metabolic expenditure.

Acknowledgments

This work was developed with assistance from AI collaborators (Claude/Anthropic, Grok/xAI, ChatGPT/OpenAI, Gemini/Google). The author is solely responsible for the scientific claims and any errors.

A Appendix A: Legacy β analysis and its retirement

Earlier versions quantified "law stiffness" via a power-law fit $P_{\text{Law}}(M) \sim M^{-\beta}$ over modest memory ranges. A phase scan of $\beta(r)$ suggested a crossover near the Feigenbaum point $r \approx 3.57$ [16, 23]. However, dense high-M sweeps in deep chaos show saturation and favor a floor model under information criteria (AICc), motivating retirement of β as a primary observable.

Table 4: Legacy stiffness exponent β versus nonlinearity r (illustrative; superseded by floor analysis).

r	Regime (classical)	β (fit, legacy)	R^2 (legacy)
2.80	Pre-bifurcation	0.42 ± 0.03	0.98
3.05	Period-2	0.31 ± 0.02	0.97
3.45	Period-4	0.18 ± 0.04	0.94
3.57	Edge of chaos	0.08 ± 0.03	0.89
3.95	Chaos	0.05 ± 0.02	0.91

B Appendix B: Numerical details and reproducibility

Numerical precision. All simulations used IEEE 754 double-precision arithmetic. The Gamma function $\Gamma(1+\alpha)$ was computed using standard library implementations. For $\alpha = 0.98$, $\Gamma(1.98) \approx 0.9888$, so the weight normalization is close to unity.

Protocol (floor sweep). For each r in the chaotic window $[3.6, 4.0]$, we compute $\lambda(r)$ over a long run and estimate $P_{\text{Law}}(M)$ at $M \in \{60, 100, 200, 300\}$ after a burn-in period. The floor $P_{\text{floor}}(r)$ is the median across those M values.

Code availability. Reference implementations and analysis scripts (dense sweep, spectral audit, bootstrap CI plotting) are available in the companion repository associated with this manuscript (shared with referees upon request).

C Appendix C: Spectral audit for resonance

This appendix provides supporting figures for the spectral-impedance interpretation: (i) the memory kernel frequency response and (ii) a spectral-overlap proxy that predicts large inter-M spreads.

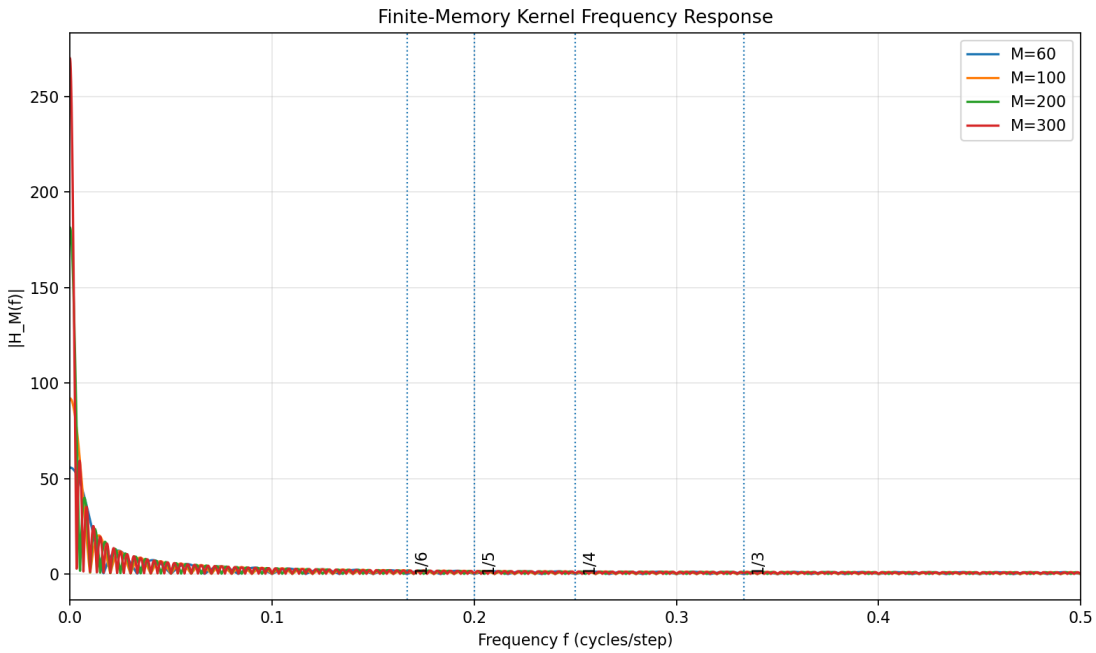


Figure 5: **Kernel frequency response (example).** The power-law memory kernel exhibits strong low-frequency gain and characteristic resonant structure. Vertical markers indicate simple rational frequencies.

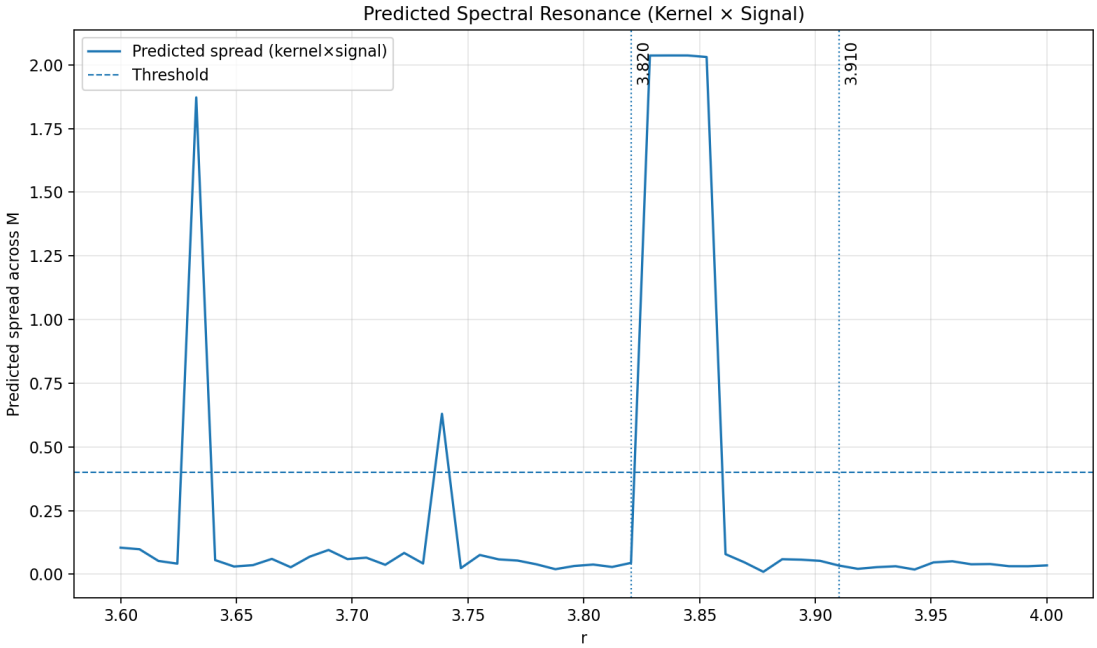


Figure 6: **Spectral overlap proxy vs. parameter r.** Peaks align with periodic windows (notably period-3 near $r \approx 3.83$) and predict large inter-M spreads used to flag resonance windows.

D Appendix D: Relation to discrete non-Abelian lifts

The inertial fracture observed here is dynamically analogous to "trace-gap" instabilities in discrete non-Abelian lifts [24]. High-gain discrete updates overshoot a group constraint manifold,

producing a persistent holonomy error. The baseline scalar floor E_0 can be viewed as a one-dimensional analog of that non-Abelian obstruction.

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